



American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

“FloodSafe - Flood Alert and Safety App”

A Software Engineering Project Submitted

By

Semester: Fall 24-25		Section: A	Group Number: 01	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
1	S.M. Tawhid	21-45470-3	18%	
2	Md Tanvir Ahamed	21-45291-2	18%	
3	Mojibul Hoque Jihad	21-45954-3	10%	
4	Fardin-al-Sezan	22-46868-1	18%	
5	Hafsanur Akter	22-48057-2	18%	
6	Md. Jamir Shekh	22-48468-3	18%	

The project will be Evaluated for the following Course Outcomes

CO3: Select appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]
Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report		[5Marks]
CO4: Develop project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Total Marks	
	Develop the project plan, its components of the proposed software products	[5Marks]
Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources		[5Marks]

Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	
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Description of Student's Contribution in the Project work

Student Name: Fardin-al-Sezan
Student ID: 22-46868-1
Contribution in Percentage (%): 18%
Contribution in the Project:

- Solution of the problem
- Functional Requirements
- Use Case diagram
- UI/UX design
- Test case
- Project estimation
- Timeline Chart 2
- Work bench structure



Signature of the Student

Student Name: S.M. Tawhid
Student ID: 21-45470-3
Contribution in Percentage (%): 18%
Contribution in the Project:

- Solution of the problem
- Functional Requirements
- Class diagram
- UI/UX design
- Test case
- Project estimation
- Timeline Chart 1



Signature of the Student

Student Name: Hafsanur Akter
Student ID: 22-48057-2
Contribution in Percentage (%): 18%

Contribution in the Project:

- Solution of the problem
- Functional Requirements
- Activity diagram
- UI/UX design
- Test case
- Sequence Diagram

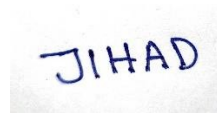


Signature of the Student

Student Name: Mojibul Hoque Jihad
Student ID: 21-45954-3
Contribution in Percentage (%): 10%

Contribution in the Project:

- Solution of the problem
- Functional Requirements
- Test case



Signature of the Student

Student Name: Md Jamir Shekh
Student ID: 22-48468-3
Contribution in Percentage (%): 18%

Contribution in the Project:

- Solution of the problem
- Functional Requirements
- Activity diagram
- UI/UX design
- Test case
- Work Bench Structure



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Signature of the Student

Student Name: Md Tanvir Ahamed
Student ID: 21-45291-2

Contribution in Percentage (%): 18%

Contribution in the Project:

- Solution of the problem
- Functional Requirements
- Sequence diagram
- UI/UX design
- Test case
- Class diagram

A handwritten signature in blue ink that reads "Tanvir Ahamed" with a horizontal line underneath.

Signature of the Student

1. PROJECT PROPOSAL

1.1 Background to the Problem

- **Background Description:**

Floods have consistently threatened lives, property and ecosystems. The rising frequency and intensity of these events, partially fueled by climate change have heightened the demand for reliable tools to predict and manage floods. Although technology and forecasting have progressed many current systems still lack timely and precise predictions.

- **Root cause problem:**

The main reason for this problem is the complex and unpredictable nature of floods. Flooding is influenced by many factors that interact in unexpected ways. For example, climate change is raising global temperatures and changing weather patterns, leading to more frequent and intense weather events. Factors like rainfall patterns, river flow, and soil moisture levels also play a role in floods and droughts, making them harder to predict accurately. On the other hand, human activities like urbanization and deforestation can increase the impact of rainfall.

Importance of Choosing this root problem:

This problem must be resolved because timely and accurate forecasts can save lives by giving people early warnings so they can take preventative action. Floods cause significant financial losses by destroying enterprises and property. The effects of floods can extend well beyond the immediate area in our society.

1.2 Solution to the Problem

- **Objective:**

The objective of the project is to develop a reliable and user-friendly mobile application that provides real-time flood alerts, safety guidance and preparedness resources. By delivering timely warnings and actionable information, the app aims to help individuals and communities mitigate risks, take preventive measures and enhance their resilience to flood-related disasters.

Solution of this problem:

Establish networks of weather stations, river flow sensors and satellite monitoring systems to continuously track rainfall, soil moisture, water levels and weather patterns. This data can be integrated to create a comprehensive view of flood risk areas. Design and build resilient infrastructure like flood barriers, drainage systems and permeable surfaces in urban areas to manage excess water more effectively. Encourage green infrastructure like wetlands and green roofs to naturally absorb rainfall and mitigate flood impact.

○ **Solutions to deal of these problem:**

- **Climate Adaptation Count:** Create policies and infrastructure to deal with the effects of climate change, such as buffer zones along rivers and coastlines, green infrastructure to absorb excess water, and flood defenses.
- **Public Awareness:** Encourage communities to use resources like emergency kits and alert apps to keep ready for flood disasters and raise public awareness of flood hazards and safety precautions.
- **Environmental Management System:** Encourage sustainable land-use techniques to lessen the effects of flooding brought on by human activities like reforestation, wetland restoration and improved urban design.

Reasons for Appropriateness of Solutions:

This solution is particularly appropriate because it directly addresses the complexities and unpredictability of flooding through a multi-faceted approach. Flood dangers are increased by human activities like urbanization and deforestation. Flood frequency and intensity are decreased by stabilizing the environment and reducing flow through the use of sustainable planning techniques, protecting natural landscapes and encouraging reforestation. Community awareness programs and early-warning systems empower individuals to act proactively when flood risks arise. This improves public safety, enabling quicker responses, and minimizing harm.

Yes, the solutions outlined for this app are feasible in meeting the business objective. Here's the reason:

The suggested fixes are realistic and in many respects in line with corporate objectives. By providing early and precise natural disaster forecasts and practical advice disaster Guard fills a vital need in public safety. The app can provide timely notifications and guidance because it makes use of cutting-edge technology including machine learning and real time data. On the other hand, it is scalable which enables it to adjust to various geographical locations and disaster kinds, promoting broad use and impact. Enhanced user engagement through personalized notifications and interactive features can draw in a sizable user base and increase money through premium choices and collaborations with governmental and emergency services. Additionally, the Disaster Guard is a powerful partner for emergency responders and municipal authorities due to its compatibility with public safety objectives.

○ **Basic functionalities:**

The water level indication is continuously monitored. Monitored Gives current information on flood including their projected times, locations and degrees of severity. Additionally, shows interactive maps with the locations of shelters, safe zones, evacuation routes and impacted areas highlighted. Includes a collection of instructional resources on how to prepare for and respond to Flood.

Deep insight on creative solution to the real-life problem:

The proposed program analysis and flood by combining AI with real-time data streams. It provides users with tailored safety suggestions depending on their location and expected

risks using predictive modelling and risk assessment. Participation of the community and crowdsourced data help to improve the predictive power of the app while real-time Critical information is promptly disseminated thanks to notifications and communication channels. The software can change over time because to methods for ongoing learning and development, which eventually enabling people and communities to anticipate and react to flood possibly saving lives and lessening their effects.

- **Target group:**

The public is among target users, particularly those who reside in flood affected area and require immediate information and guidance. It also covers local governments and NGO to assist Flood management plans and public safety initiatives, emergency responders who need a tool for better coordination and information dissemination and employee safety.

- **Contributions to the development of Scientific Results:**

Flood advances predictive analytics and integrates interdisciplinary data sources to help generate scientific findings. It will combine information from multiple sources like social media feeds, satellite photography, geological data, and meteorological data, offering a thorough analysis of possible flood situations. Working along with meteorologists, Interdisciplinary research is fostered by geologists, data scientists, and emergency management specialists and inventiveness. The application will employ cutting-edge AI and machine learning algorithms to forecast floods. Analytics in real time will show notable developments in the processing of real-time data from and user inputs, supporting scientific investigations into the ways in which real-time data might enhance prediction accuracy and promptness.

- **Literature Review on Existing Studies:**

Significant research has been done on the Flood Alert and Safety App, with several studies offering insights into the variables influencing flood risk and strategies to improve prediction accuracy. For example, **Chen et al. (2018)** demonstrated how integrating data from weather sensors and river levels greatly enhances forecasts by presenting an integrated strategy that uses machine learning to estimate flood hazards based on real-time data. Similar to this, **Jones and Wu (2021)** stressed the significance of integrating soil moisture levels and rainfall data to provide early flood warnings, pointing out that human activities like urbanisation and deforestation exacerbate the effects of flooding, especially in metropolitan areas. The impact of climate change on flood risks was examined in another study by **Lee and Kumar (2020)**, which came to the conclusion that increasingly severe flood events are directly associated with rising global temperatures.

- **Existing Studies and Software Solutions:**

A solid basis for comprehending flood dangers and enhancing early warning systems is provided by the existing research on Flood Alert and Safety App. While McCloskey et al. (2016) suggested using AI for real-time rainfall and soil moisture analysis, Smith et al.

(2019) emphasised the potential of merging satellite data with ground-based sensor readings to boost flood forecast accuracy. By providing timely warnings and risk assessments, these studies highlight the importance of data-driven models in reducing the effects of flooding. Based on these discoveries, a number of software solutions have been created. For instance, FloodMap Mobile uses satellite technology to provide localised flood risk information, but with limited geographic coverage. By developing a mobile application that combines many data sources for a comprehensive, real-time view of flood threats, our project, the Flood Prediction and Management System, broadens the scope of these solutions. Our app improves public safety and individual preparedness by providing interactive maps that show safe zones, shelters, and evacuation routes in addition to tracking rainfall and flood levels.

○ **Existing Software Solutions and Proposed Solution Extends User Benefits:**

Existing flood prediction software programs, such RiverMon and FloodMap Mobile, have restricted coverage and frequently only concentrate on particular areas or data sources. While My Shake and QuakeAlertUSA focus on earthquake notifications, AccuWeather and The Weather Channel offer broader weather forecasts. Although these resources are useful for raising awareness generally, they fall short in providing thorough, flood-specific advice and safety suggestions. By integrating several data sources, including satellite feeds and weather stations, into a single mobile app platform, our suggested Flood Alert and Safety App fills in these gaps. This method allows the app to provide interactive maps that show nearby shelters, safe zones, and suggested evacuation routes in addition to more accurate and fast flood warnings. Additionally, our app uses community-sourced data and AI-driven predictive modelling to improve accuracy, offering location-specific safety recommendations that adjust to the needs of the user. Our system provides users with an all-in-one tool that not only gets them ready for floods but also gives them actionable insights specific to their local surroundings by combining real-time data and AI algorithms. This comprehensive strategy closes a significant gap in the existing environment, ultimately improving user safety.

2. Requirements Analysis:

1. Software Sign up :

Functional Requirements:

- For new users the app will offer a signup page.
- Username, E-mail and password text fields will be included on the signup page.
- The application will confirm that the selected E-mail address and username are valid.

- If the specified E-mail address or username is already in the database there will be One popup message instructed me to enter a different email address or username.
- The user's information will be safely saved in the database after a successful signup.

Priority Level: Medium

Preconditions: E-mail address and username are must be unique.

2. Software Sign in:

Functional Requirements:

- Existing users will have access to a sign-in page through the app.
- Text fields for the username and password must be present on the sign-in page.
- The application will check the database to make sure the information entered are correct.
- The user will be taken to the app's main screen after successfully logging in.

Priority Level: Medium

Preconditions: User must have a registered account.

3. Location and Notification Access:

Functional Requirements:

- The application will ask for authorization to access the user's location like the app might request access to the location so it can inform the user if they are in a flood warning area or provide evacuation routes in case of an imminent flood. This request would typically appear as a prompt on the user's device, explaining the necessity of location access to offer the service and the user would have the option to grant or deny permission.
- In order to deliver push notifications the app must ask for permission.
- A popup message box asking for the location and notifications will appear the user's access.
- To interact with this page a Allow/Don't allow button will be there.

Priority Level: High

Preconditions: Device must have location and notification services enabled.

4. Continue without sign in/sign up:

Functional Requirements:

- Users will have the option to "continue without sign in/sign up" using the a
- By selecting the "continue without sign in/sign up" option then users can move straight to the app's home screen.

Priority Level : Medium

Precondition : Location and notification access have already been granted by users.

5. Forgot Password:

Functional Requirements:

- On the sign-in page, the app will offer a "Forgot Password" option.
- When users forget their password they can retrieve their account on a different page.
- An email address entry prompt will appear for the user.
- The application will email the user a link to reset their password.
- To restore their data users must click on the link that was supplied to their email address.

Priority Level : Medium

Precondition : User must have a registered email address .

6. Reset Password:

Functional Requirements:

- The app shall provide a page for users to reset their password. The password reset process would likely involve the user providing their registered email address or phone number and then receiving code to reset the password. This functionality is essential to maintain the security and accessibility of user accounts, especially for an app like FloodSafe.
- The page shall include fields for entering a new password and confirming it.
- The app shall validate the new password and update it in the database upon submission

Priority Level : Medium

Precondition : User must have received a password reset link on E-mail

7. Real-Time Temperature Monitoring:

Functional Requirements:

- The application will show the user's present location's temperature in real time.
- The GPS system will be used to retrieve the current location.
- The application will periodically update temperature data.
- The app's home screen will showcase this feature.
- Users can also see the weather and temperature in other cities and places.

Priority Level : High

Precondition : User must have granted location access.

8. Flood Alerts and Forecast:

Functional Requirements:

- Using real-time data, the app will send out alerts for flood
- The color of the particular widget or container that shows the notifications will be altered based on the level of emergency.
- Forecasts for impending natural disasters will be shown on the app.
- The alerts will include details about the kind of crisis that may occur, the includes the alert title as well as the disaster's intensity level.

Priority Level : High

Precondition : User must have granted location access

9. Location-Based Alerts:

Functional Requirements:

- Alerts tailored to the user's current location will be sent by the app.
- This will be an in-app alert rather than a notification alert.
- The app will have a specific alert text or option.
- By selecting the alert option, users can view current and upcoming alerts based on their present position.
- To ascertain the user's location, the app will employ GPS technology.

Priority Level : High

Precondition : User must have granted location access

10. Safety Guidelines and Procedure:

Functional Requirements:

- The app shall provide safety guidelines and procedures for flood
- It will provide necessary guidelines and precautions needed for flood
- The guidelines shall include evacuation routes, emergency contacts and safety tips.

Priority Level : High

Precondition : None

11. User Reporting in Emergency:

Functional Requirements:

• In the FloodSafe app, users will have the ability to report any emergencies or hazards they come across, enabling them to quickly alert authorities and others in the area for prompt action and response.

- There will be a option of 'report emergency' inside the app. Users can click on the option and a new page will be open. User can there write the emergency they faced or facing and submit to the app.
- The app shall provide a option and a form for users to submit relevant information.

Priority Level : High

Precondition : None

12. Regular Notifications alerts:

Functional Requirements:

- Regular notifications for significant updates and reminders will be sent by the app.
- The software will notify users of weather forecasts and updates on a regular basis.
- Notifications will be sent to users at certain times, such as in the morning.
- In the settings, users will be able to choose whether to allow or disable these notifications.

Priority Level : High

Precondition : User must have granted notification access.

13. Emergency Notifications with Guidelines:

Functional Requirements:

- The app will notify the user of emergencies and provide pertinent safety instructions. This means that this feature ensures that users receive real-time updates and practical guidance, which is critical for their safety during natural disasters like floods.
- The recommendations must be tailored to the emergency's nature and level of severity.
- Users have the choice to enable or stop this notification feature once more.

Priority Level : High

Precondition : None

14. Rescue Team in Emergency situation upon users request:

Functional Requirements:

- Following that, the Drone will locate by using Image processing object detection to victim user and proceed to rescue them.
- When user requests emergency assistance the app will dispatch an emergency rescue team to them.
- The application will use real-time data to validate the user's request.

Priority Level : High

Precondition : Camera access and Storage access on device.

15. Real time map with accurate data of weather and natural disaster:

Functional Requirements:

- The app will feature a real-time map that shows the global weather conditions.
- Within the map, users may view the current weather conditions.
Users can view and remain informed about impending severe weather conditions by using a real-time map.

Priority Level : High

Precondition : Location access must be granted by the user.

16. Offline Mode for Critical Information:

Functional Requirements:

- To enable users to access them even when offline, the app will keep important safety instructions and information offline.
- Users will be able to receive vital safety advice, escape routes, and emergency contacts that aren't online.
- When the app is online, it will periodically sync data to make sure the data is current.

Priority Level : High

Precondition : Required data must first be downloaded by users during they are online

17. Feedback and Reporting from the Community:

Functional Requirements:

- Users will be able to report on-ground conditions and offer feedback using the app. This feature helps create a community-driven flow of information, where users can share firsthand reports, which could aid others in the area or help authorities assess the situation more accurately
- Users have the option to submit images, videos and descriptions of the local environment which will be shared with emergency services and other users.
- To give users real-time updates on local events the app will show these reports on device.

Priority Level : Medium

Precondition : Camera access and Storage access on device.

18. Emergency Contact List:

Functional Requirements:

- Users will be able to make and manage an emergency contacts list through the app.
- In the event of an emergency users can promptly notify their contacts via phone or messages

Priority Level : Medium

Precondition : Users must save contact information within the ap

19. Flood area prediction Checklists:

Functional Requirements:

- The app will give users checklists to help them get ready for flood
- These checklists can be altered by users to suit their own requirements and situations.
- The software will remind users to do preparation chores by the due dates approach.

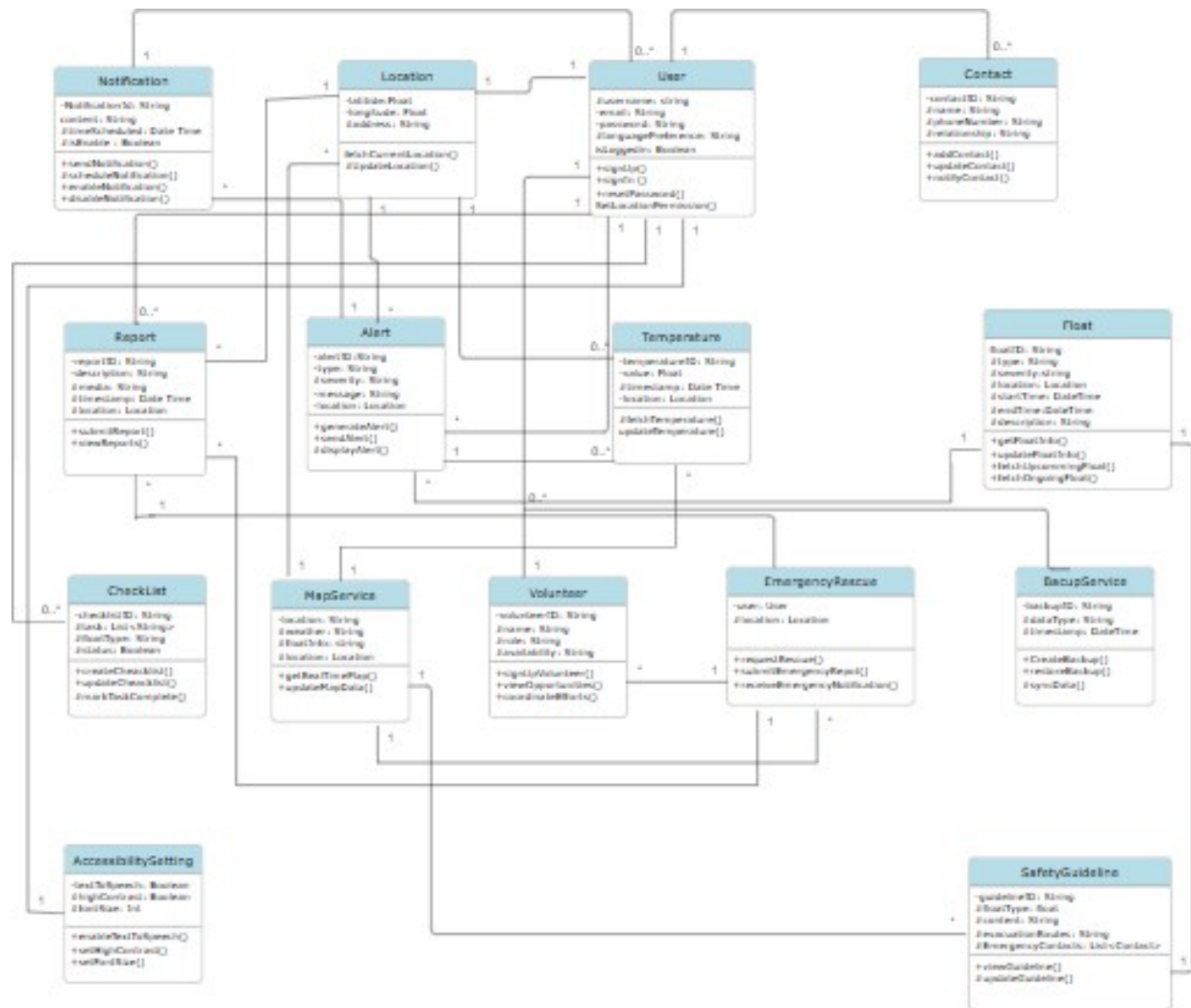
Priority Level : Medium

Precondition : None

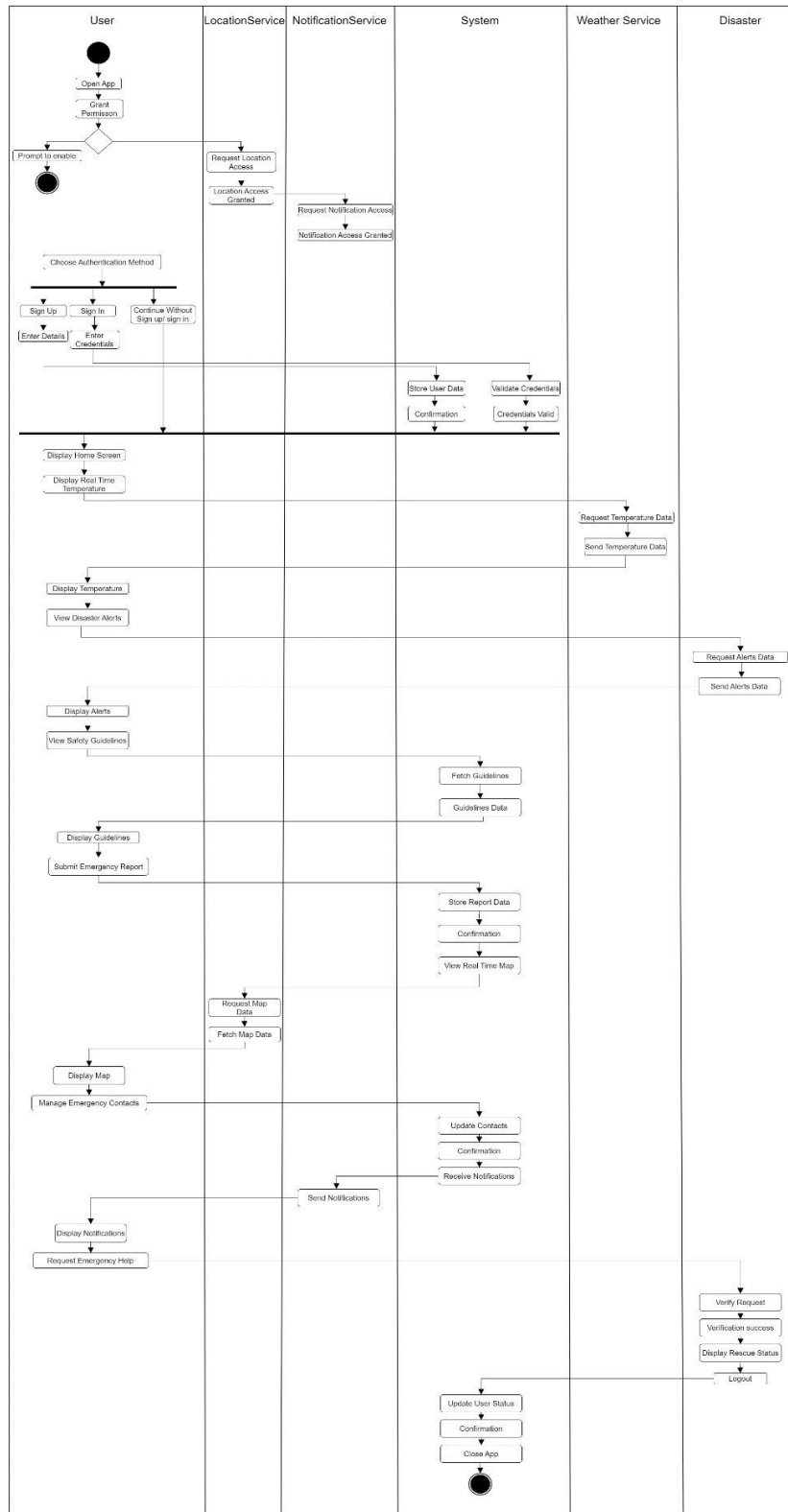
Use Case Diagram:



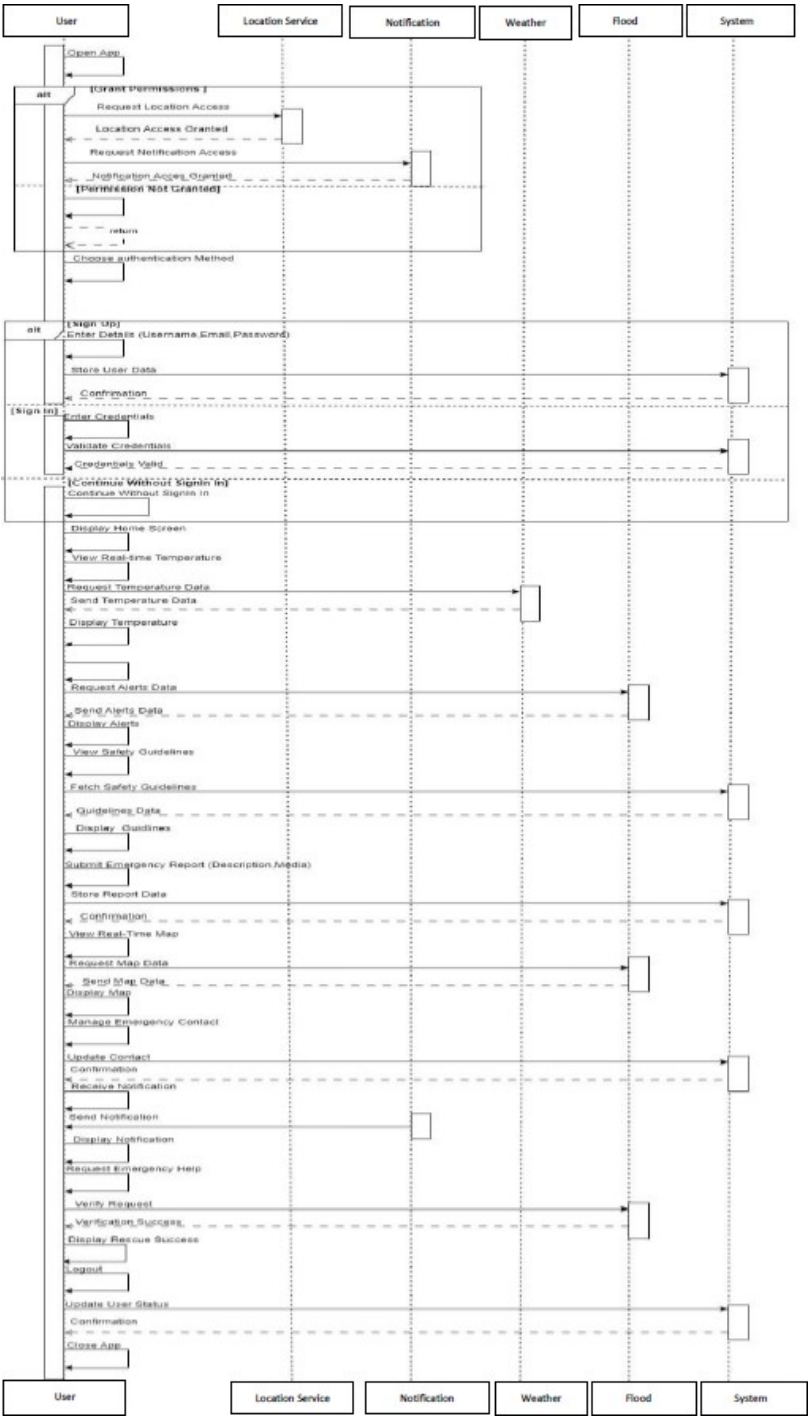
Class Diagram:



Activity Diagram:



Sequence Diagram:



3. SOFTWARE DEVELOPMENT LIFE CYCLE

3.1 Software Process Model

Scrum is the best option for creating Flood because it is flexible and adaptable, two qualities that are critical for an application that has to react rapidly to changing user requirements and new disaster scenarios. Scrum's iterative methodology enables quick modifications, guaranteeing the app stays useful and relevant as Situations shift. Furthermore, the regular releases made possible by Scrum's brief sprints guarantee that the software is regularly updated with the most recent safety instructions and disaster information, giving users with up-to-date, trustworthy facts because Scrum integrates user feedback throughout the development cycle, this frequent update method guarantees that the software develops in accordance with user expectations. The focus on user-centered development aids in improving the usability and functioning of the app, increasing its efficacy in practical uses.

Additionally, creating Flood necessitates smooth collaboration between end users, developers, and specialists in disaster management. Scrum's emphasis on frequent communication and teamwork guarantees that all parties involved are on the same page and pursuing the same objectives, improving the overall quality and efficacy of the application. Additionally, Scrum's iterative process makes ongoing risk management easier. Enabling the group to spot possible problems early and fix them, guaranteeing the program can manage unforeseen circumstances successfully.

Scrum's focus on quality assurance and frequent testing helps to maintain high standards by making sure every release is dependable and strong, which is essential for an app utilized in emergency situations. Finally, Scrum's prioritization of the product backlog guarantees that the most Prioritizing the development and delivery of valuable products maximizes user benefits and guarantees that Limited resources are used effectively.

Why Not other process Models?

Since Scrum offers a higher degree of flexibility, adaptability, and customer-centric focus than alternative models like Waterfall, V-Model, Incremental, or XP, we are not adopting them. The strict, linear methodology of the V-Model and Waterfall models makes it difficult to adapt to changes once the project is underway, causing They are less appropriate for applications like the Flood app that have changing requirements. Gradual Although more adaptable, development places less of an emphasis on iterative learning and feedback as Scrum. Although XP emphasizes technical excellence, it does not have the Scrum-provided formal project management framework. Scrum's iterative cycles, regular evaluation, and focus on user feedback guarantee that the final product stays in line with user needs and can swiftly adjust to new information or environmental changes, which makes it the ideal option for intricate and dynamic applications like Flood.

3.2 Project Role Identification and Responsibilities

Product Owner

The success of the product is mostly the responsibility of the product owner.

Responsibilities

The product backlog is defined and prioritised by the product owner, who also makes sure that it represents stakeholder and user demands. They inform the development team of the project's vision and, in response to input and business goals, choose features and priorities. Being the main point of contact for interested parties, they make sure Throughout the project, requirements are fulfilled.

Scrum Master

The Scrum Master ensures the team follows Scrum practices and removes obstacles.

Responsibilities:

The Scrum Master leads the team in adhering to Scrum principles, facilitates Scrum rituals, and promotes seamless workflow by removing obstacles. To create a cooperative and effective atmosphere, they provide the team with coaching on self-organization and ongoing development.

Scrum Team

In order to accomplish sprint goals, the Development Team works together to build the product increment.

Responsibilities:

During each sprint, they are responsible for designing, creating, testing, and delivering product increments while guaranteeing that the final products are of the highest caliber. To decide how to effectively accomplish goals and overcome obstacles, the team self-organizes. They take an active role in sprint planning retrospectives, and evaluations to consistently improve the development process and fulfil the project's goals.

Customers

The customers of the Flood app play a pivotal role in its development and success.

Responsibilities:

Clients participate in user testing, supply precise specifications, and offer input at every stage of development. They assist in setting feature priorities according to the app's use in disaster management and other important requirements.

Their participation guarantees that the software satisfies practical requirements and improves its relevance and usability.

Management

The management team overseeing the development of the Flood app holds several critical responsibilities.

Responsibilities:

Management communicates with stakeholders, assigns resources, and establishes clear goals. To guarantee that the app satisfies user expectations and quality requirements, they supervise risk management and offer strategic direction. Management facilitates the effective implementation of the Flood by encouraging teamwork app as a useful instrument for disaster management.

The Flood app may be created quickly and effectively by utilizing the Scrum framework and responsibilities that are well-defined. This will guarantee that a high-quality solution that satisfies customers' changing needs during catastrophes is delivered on time.

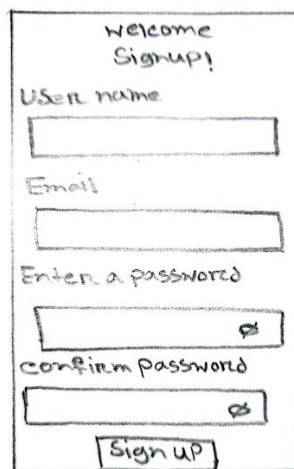
Socio Economic Effect of Flood

The Flood app enhances preparedness and reaction to disasters, which has major socioeconomic benefits. Socially, it makes communities more resilient by facilitating prompt coordination and communication in times of crisis, which makes users feel safer and more united. In terms of finances, it lowers the financial strain on both individuals and governments by reducing harm through prompt notifications and precautions. By expediting the recovery process and decreasing downtime, it promotes economic stability in areas vulnerable to disasters. In general, Flood enhances social safety nets and encourages economic resilience by reducing losses and making it possible to allocate resources effectively in times of crisis.

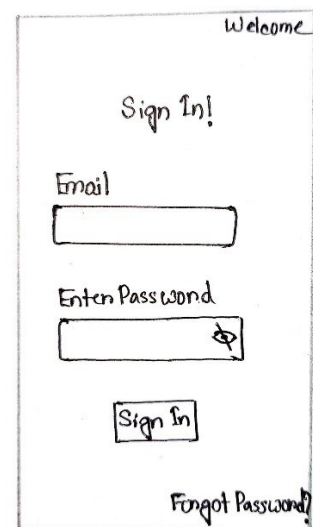
UI Design:



Initial page



Sign up page



Sign in Page

Welcome

Recover Password

Enter Your Email

Submit

Sign In

Password Recovery page

Location

Temperature

13°C

Hourly Forecast

Time	Time	Time	Time
Temperature	Temperature	Temperature	Temperature

Tomorrow

Temperature

Home page 1

Location

Temperature

12°C

Danger!

Possible Flood Alert

Hourly Forecast

Time	Time	Time	Time
Temperature	Temperature	Temperature	Temperature

Tomorrow

Temperature

Home page 2

Flood

Possibility
noise water

Date & Time

Don't panic :)

Follow the Safety tips -

- * Safety tips 1
- * Safety tips 2
- * Safety tips 3

Disaster Page

Flood

Possibility
noise water

Date & Time

Don't panic :)

Follow the safety tips -

- * Safety tips 1
- * Safety tips 2
- * Safety tips 3

Menu Bar

Temperature

Pressure

Safety Guideline

Settings

Menu Bar

Location

VIEW CONTACT FOR SPECIFIC LOCATION

Select Location

Location

Contact List

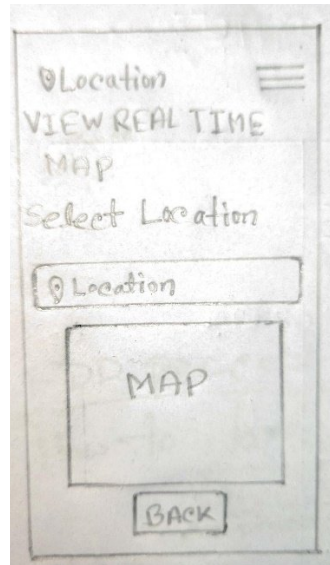
1. Contact 1

2.

3.

Flood Comment

Contact List page



Real time map Page

Test plan: The test plan for the Flood app involves conducting functionality testing to ensure that each feature operates as expected. The testing will cover the eight identified functional requirements, including real-time temperature monitoring, disaster alerts, safety guidelines, user reporting, emergency rescue, contact lists checklists, and volunteer coordination. Functionality testing will be conducted by a team consisting of a Lead Tester, two Senior Testers, and three Junior Testers. Each tester will be responsible for specific features, ensuring comprehensive coverage. The testing will be implemented using unit tests for individual components, system testing to verify the app's overall functionality, and integration testing to ensure that different parts of the system work together seamlessly. After functionality testing, acceptance testing will be conducted to validate that the app meets the end-users' needs and business requirements. This phase will involve key stakeholders and will be carried out in the last week, ensuring that the app is ready for deployment. Functionality testing will take place during the first two weeks, followed by acceptance testing in the final week.

Test Case:

1

Project Name: FloodSafe - Flood Alert and Safety App			Test Designed by: FARDIN-AL-SEZAN	
Test Case ID: FR_08			Test Designed date :15-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Rescue Team in Emergency situation upon user’s request			Test Execution date:	
Test Title: The app shall send emergency rescue team to the user upon users request for emergency help				
Description: Test the app’s ability to send a rescue team to the user's location upon request and track the user's real-time location.				
Precondition (If any): The user must have given location access to the FloodSafe app				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Open the application 2. Go to the Rescue Team page 3. Enter Request Type 4. Location will be auto fetched of the user 5. Click on ‘Submit’	Request Type: "Rescue Emergency"; User Location: Bashundhara, Dhaka.	After click on the ‘Submit’ button the app shall receive the rescue request and send team to that specific location for rescue the user		
Post Condition: User is successfully rescued from the emergency location and reached at a safe place.				

Project Name: FloodSafe - Flood Alert and Safety App			Test Designed by: MD. JAMIR SHEKH	
Test Case ID: JS_09			Test Designed date :15-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Real-Time Temperature Monitoring			Test Execution date:	
Test Title: The app shall display real-time temperature data for the user's current location.				
Description: Verify that the app displays accurate real-time temperature data for the user's current location and allows viewing temperatures for other locations.				
Precondition (If any): The user must have allowed access location and GPS enabled				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Open the application 2. By default, app will be redirected to the ‘Home’ page 3. If not, go to the ‘Home’ page	Location: "Bashundhara" GPS Enabled: "Yes".	The app should display the current temperature for New York, NY, and allow viewing temperatures for other cities.		
Post Condition:				

03

Project Name: FloodSafe - Flood Alert and Safety App		Test Designed by: S.M. TAWHID		
Test Case ID: SM_07		Test Designed date :15-12-2024		
Test Priority (Low, Medium, High): High		Test Executed by:		
Module Name: Emergency Contact List		Test Execution date:		
Test Title: The app shall allow users to create and manage a list of emergency contacts.				
Description: Ensure that the app allows users to manage a list of emergency contacts and notifies them when the user is in a flood zone area.				
Precondition (If any):				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Open the application 2. Go to the 'Contact List' page 3. Add or edit contact list	Contact Name: "S.M. Tawhid" Contact Number: "01709006947"	The app should add the contact to the emergency list and notify them if the user is in flood zone area		
Post Condition:				

04

Project Name: FloodSafe - Flood Alert and Safety App			Test Designed by: MOJIBUL HOQUE JIHAD	
Test Case ID: MH_10			Test Designed date :15-12-2024	
Test Priority (Low, Medium, High): Medium			Test Executed by:	
Module Name: Flood area prediction checklists			Test Execution date:	
Test Title: The app shall provide checklists for users to prepare for various types of floods.				
Description: Verify that the app provides customizable checklists for flood prediction and sends reminders for completing tasks.				
Precondition (If any): User should log in to the application using userid and password to save their checklists in the database				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Open the application 2. Go to the ‘Checklist’ page 3. Edit and manage the checklist	Checklist Type: "Feni Flood Preparedness". Task: "Stock up on water".	The app should allow the user to customize the checklist and send reminders for completing the tasks.		
Post Condition:				

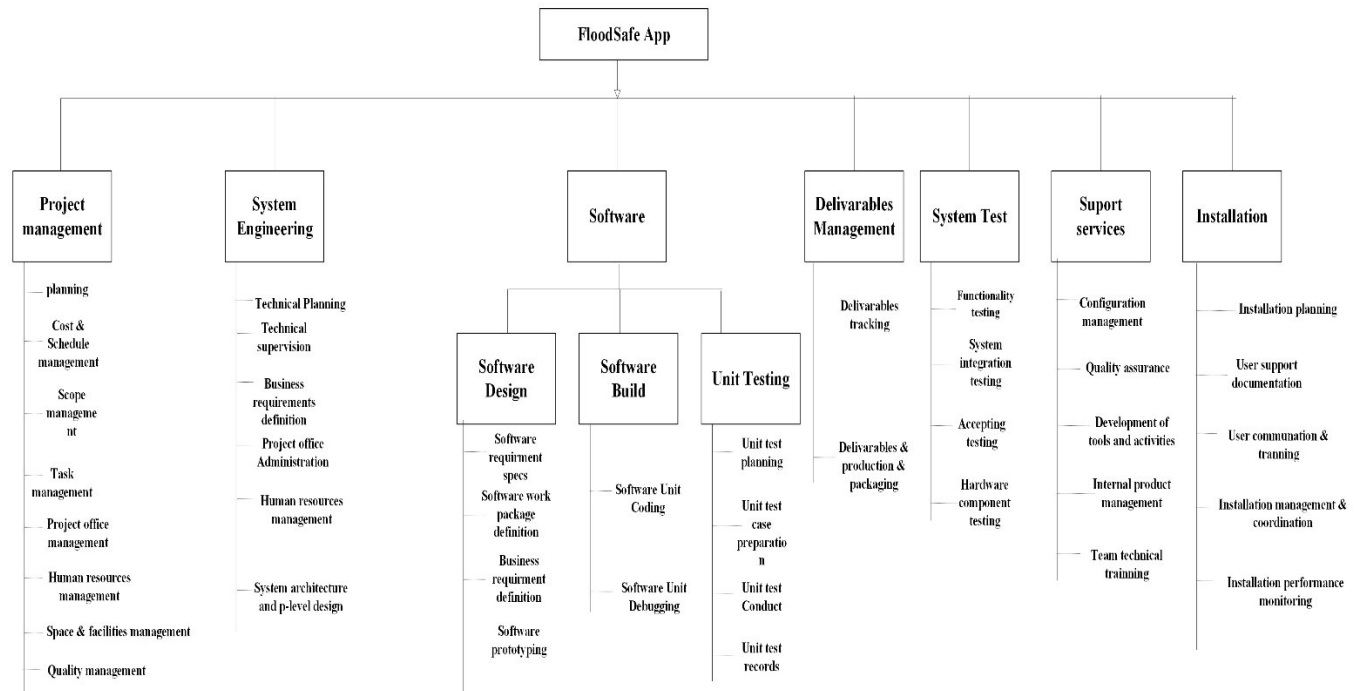
Project Name: FloodSafe - Flood Alert and Safety App			Test Designed by: MD. TANVIR AHAMED	
Test Case ID: TA_11			Test Designed date: 15-12-2024	
Test Priority (Low, Medium, High): High			Test Executed by:	
Module Name: Flood Alerts and Forecast			Test Execution date:	
Test Title: Flood Alert Notification Accuracy Test				
Description: Verifies that flood alerts are accurate, timely, and correctly delivered to users.				
Precondition (If any): The system must have access to current and accurate flood data to generate and test alerts.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Open the Application 2. Allow to access location 3. Check notification	Location: Bashundhara, Dhaka	User receives the alert notification through the email or SMS within the expected timeframe.		
Post Condition: The user is notified of the relevant information according to their location.				

Project Name : FloodSafe - Flood Alert and Safety App		Test Designed by : HAFSANUR AKTER		
Test Case ID : HA-23		Test Designed date : 16-12-2024		
Test Priority(Low, Medium, High) : High		Test Executed by :		
Module Name : Volunteer Coordination and Support		Test Execution date :		
Test Title :Volunteer Availability Update				
Description : Test the app’s feature for coordinating volunteer efforts, allowing user to sign up for opportunities and communicate with organizations				
Precondition (If any) : The volunteer is already registered and has an active account in the system.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Log Into the application 2. Navigate to the ‘volunteer’ page 3. Sign up as a ‘volunteer’ 4. Manage and check availability and accessibility update		Volunteer Role : ‘Food Distribution’ Availability : ‘Weekends’		The app should allow the user to sign up for the role and receive on volunteer needs
Post Condition				

Project Name : FloodSafe - Flood Alert and Safety App		Test Designed by : FARDIN-AL-SEZAN		
Test Case ID : FAS_11		Test Designed date :15-12-2024		
Test Priority(Low, Medium, High) : Medium		Test Executed by :		
Module Name : User Reporting in Emergency		Test Execution date :		
Test Title : Verify Emergency Report Accuracy				
Description : Confirm that the information submitted by users in emergency reports is accurately captured and reflected in the system.				
Precondition (If any) :				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Check 'Report Emergency' is present. 2. Click and confirm the reporting page loads. 3. Verify form fields are correct and data is processed. 4. Simulate errors and verify app responses.	Emergency Type: ‘Flood in downtown area’ Media : ‘Attached Image’		The app should allow user to submit the emergency report successfully	
Post Condition				

Project Name : FloodSafe - Flood Alert and Safety App		Test Designed by : MD. JAMIR SHEKH		
Test Case ID : JS_10		Test Designed date :15-12-2024		
Test Priority(Low, Medium, High) : High		Test Executed by :		
Module Name : Safety Guidelines and Procedure		Test Execution date :		
Test Title: Safety Protocols and Procedures availability				
Description : The guidelines shall include evacuation routes, emergency contacts, and safety tips.				
Precondition (If any) : The emergency reporting system must be fully operational and accessible to users.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass / Fail)
1. Open the FloodSafe app and go to "Safety Guidelines." 2. Verify that safety guidelines include emergency contacts, 3. Ensure the displayed info is accurate, clear, and free from errors.	User location: Bashundhara, Dhaka		The app displays the correct and relevant safety guidelines for flood zone in the user's location	
Post Condition :				

Work Bench structure



Project Estimation

Project Type: Organic

Were,

$$P = 1.05$$

$$T = 0.38$$

$$\text{Coefficient} = 2.4$$

$$\text{SLOC} = 6000$$

Now,

$$\text{PM} = \text{Coefficient} \times \text{Effort Factor} * (\text{SLOC} / 1000) ^ P$$

$$= 2.4 * (6000/1000) ^ 1.05$$

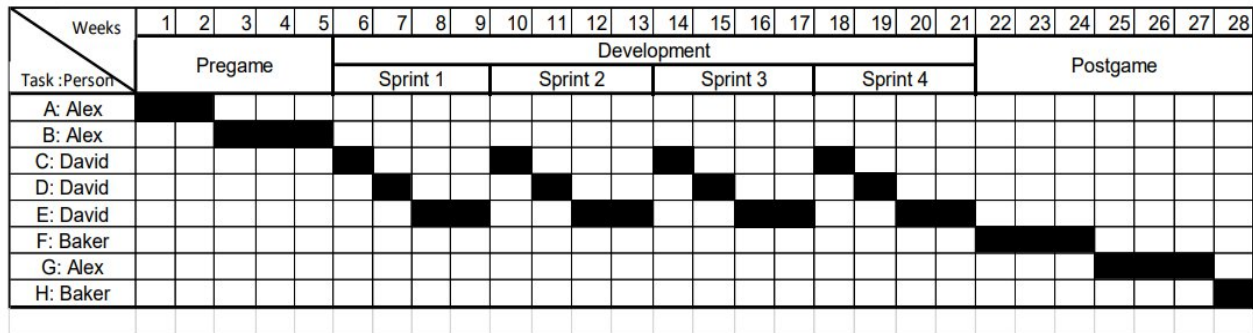
$$= 15.75$$

$$\text{DM} = 2.50 * (15.75) ^ 0.38$$

$$= 7.127 \approx 7$$

$$\text{Required number of people} = \text{ST} = \text{PM} / \text{DM} = 15.75/7 = 2.25 \approx 3$$

Timeline chart



A: Planning - Defining project goals, scope, requirements, and creating a project plan.

B: Setup Environment - Configuring development and testing environments.

C: Analysis - Studying user needs, requirements, and potential challenges.

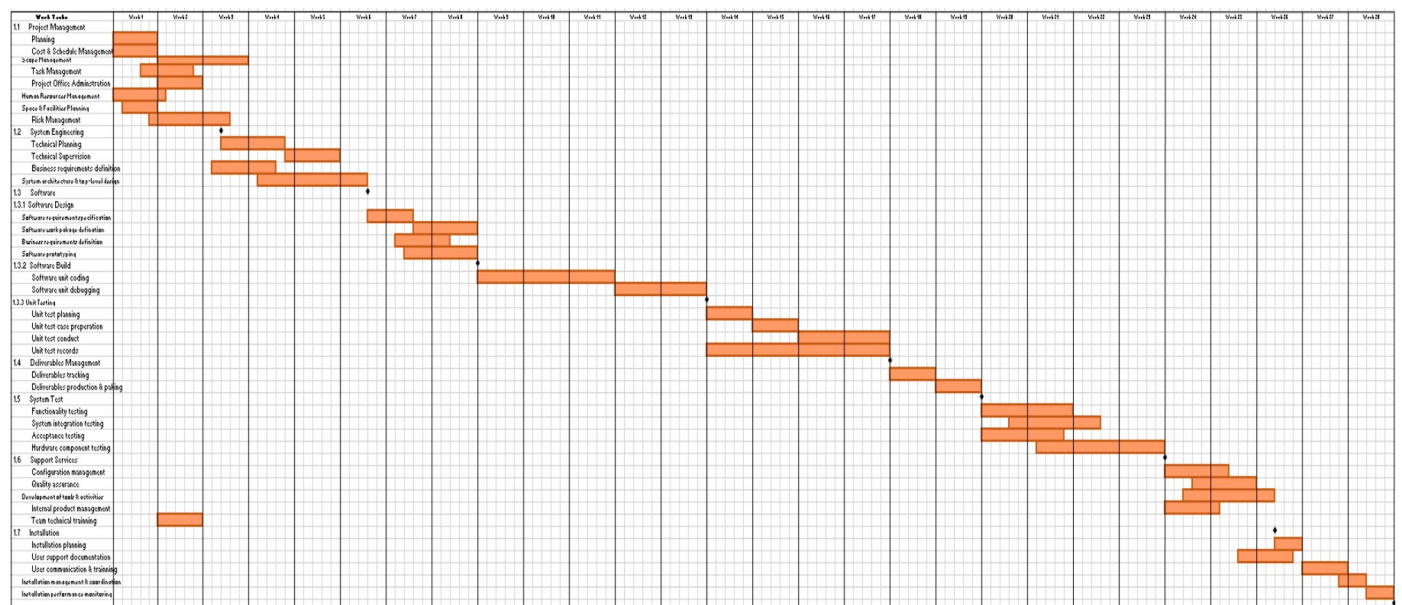
D: Coding - Writing and implementing the actual code for the system.

E: Design - Creating the user interface (UI) and visual design of the system.

F: Integration Testing - Testing how different components of the system work together.

G: System Testing - Evaluating the entire system to ensure it meets objectives and performs as expected.

H: Acceptance testing - Validating the system against user requirements to ensure it meets the agreed-upon criteria before final deployment.



Earn value analysis :

Effort estimated = $15.75 * 22$

= 346.5 person-day

Total task = 43

10 tasks have been completed but the project schedule indicates that 15 tasks should have been completed in that time.

Task	Planned Effort	Actual Effort
1	11.0	12.5
2	16.0	15.5
3	6.0	7.0
4	12.0	10.5
5	11.0	12.0
6	5.0	6.0
7	6.0	5.0
8	14.0	14.0
9	11.0	11.0
10	15.0	14.5
11	6.0	-
12	8.0	-
13	5.0	-
14	4.0	-
15	6.0	-

BCWP= 107

BCWS= 137

ACWP=108

Here,

BAC = 346.5

BCWS = 136

BCWP = 107

ACWP = 108

$SPI = BCWP / BCWS = 107/136 = 0.7868$

$SV = BCWP - BCWS = 107-136 = -29$

person-day $CPI = BCWP / ACWP =$

$107/ 108 = 0.99$

$CV = BCWP - ACWP = 107 - 108 = -1$ person-day

% Schedule for completion = BCWS / BAC

= 136 /346.5

= 39.25% [% of work schedule to be done at this time]

% Complete = BCWP / BAC

= 107 /346.5

= 30.88% [% of work completed at this time]

Risk Estimation:

Risks	Category	Probability	Impact	RMMM
Size estimation maybe significantly low	PS	60%	2	Implement regular progress reviews
larger number of users than planned	PS	30%	3	Design the system with scalability in mind to accommodate potential increases in user volume
less reuse than planned	PS	70%	2	Develop user-centric features and functionalities based on feedback and user testing
End-users resist system	BU	40%	3	Conduct deep testing, including stress testing and scenario-based testing, to identify and address potential technical glitches
Delivery deadline will be tightened	BU	50%	2	Regularly monitor project progress and adjust resource allocation as needed to meet delivery deadlines
Funding will be lost	CU	40%	1	Diversify funding sources and secure alternative funding options

				to mitigate the risk of funding loss
Customer will change requirements	PS	80%	2	Communicate effectively with stakeholders to manage expectations and negotiate changes within project constraints
Technology will not meet expectations	TE	30%	1	Conduct thorough technology evaluations to
Lack of training on tools	DE	80%	3	Provide training and support for team members to ensure this doesn't happen
Staff inexperienced	ST	30%	2	Implement strategies to improve employee retention, such as offering competitive compensation and benefits packages
Staff turnover will high	ST	60%	2	Implement strategies to improve employee retention, such as offering competitive compensation and benefits packages
Delayed rescue team response due to network issues	TE	50%	2	Can sometimes response late cause of network issue
Inaccurate disaster alerts due to faulty data sources	TE	40%	3	Conduct thorough requirements validation sessions with stakeholders to ensure a clear understanding of user needs and expectations
System crashes during peak user traffic	PS	50%	3	Can sometimes app crashes because of network issue

Impact Values

- 1 – catastrophic
- 2 – critical
- 3 – marginal
- 4 – negligible

Category Notations

- PS – Product size
- BU – Business Impact
- CU – Customer Characteristics
- TE – Technology to be built
- DE – Development environment
- ST – Staff size and Experience

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing	Some errors in spelling and grammar. Some problems	Few errors in spelling and grammar. Presents most of the details in	Project report is complete and No errors in spelling and grammar. Consistently	

	organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	a logical flow of organization in definition, details, and example.	presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO4)

Marking Criteria	Marks Distribution (Maximum 3X5=15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					